NETFRAME

Homelab Infrastructure Runbook

km-cluster | NetFRAME CS9000 42U

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0 Contents

1 Infrastructure Overview

NetFRAME is a professional-grade homelab built inside a NetFRAME CS9000 42U rack, purpose-built as a live CCNA lab, LLM inference platform, ML research node, and backup infrastructure. The cluster is named **km-cluster** and runs Proxmox VE 9.2.3 across seven nodes as of July 2026 (Randy stayed on 9.1.1 — a kernel/ZFS-only upgrade).

1.1 Cluster Nodes

Hostname	Role	IP	CPU	RAM	Notes
Randy	Storage / PBS	192.168.10.187	2x E5-2690 v3	128 GB	SuperMicro CSE-219U
QuarkyLab	ML / research	192.168.10.179	2x E5-2699 v4	512 GB	R730, RTX 8000 48GB
Jarvis	LLM Inference	192.168.10.31	2x E5-2687W v4	384 GB	R730, 2x RTX 6000 (48GB)
pve2	OPNsense host	192.168.10.204	i7-8700	32 GB	HP EliteDesk G4 SFF
pve3	Cluster node	192.168.10.201	i7-8700	48 GB	HP EliteDesk G4 SFF
pve4	Cluster node	192.168.10.202	i5-7500T	32 GB	HP EliteDesk G3 Mini
pve5	Cluster node	192.168.10.203	i5-7500T	32 GB	HP EliteDesk G3 Mini

1.2 Management IPs

Device	IP
EX3400 (switch)	192.168.10.50
OPNsense LAN	192.168.10.1
Randy IPMI	192.168.20.22 (VLAN 20)
QuarkyLab iDRAC	192.168.20.20 (VLAN 20)
Jarvis iDRAC	192.168.20.21 (VLAN 20)
Pi-hole / DNS	192.168.10.177
Ares (admin)	192.168.10.100 (wired) / .199 (WiFi)

2 Network Architecture

2.1 VLAN Design

VLAN	Name	Subnet	Notes
1	mgmt	192.168.10.0/24	Management, all servers
20	trusted	192.168.20.0/24	Trusted devices
30	servers	192.168.30.0/24	Service VMs
40	IoT	192.168.40.0/24	IoT devices
50	VoIP	192.168.50.0/24	Cisco 8841 phones
60	guest	192.168.60.0/24	Guest WiFi
70	lab	192.168.70.0/24	CCNA lab segment

2.2 Switching

- **Juniper EX3400-48P** — enterprise fabric, JunOS 23.4R2-S7.4, IP 192.168.10.50

- **UniFi Switch 24 PRO (PoE+)** — consumer fabric (IoT, VoIP, guest)
- Inter-switch uplink: ge-0/0/32 (copper RJ45, active); xe-0/2/3 DAC (link-down, speed mismatch)

2.3 10G Fabric — ConnectX-3 DAC Links

Node	NIC Port	EX3400 Port
Randy (nic3)	ConnectX-3	xe-0/2/0 — Up 10G (VLAN 30)
Jarvis (enp132s0)	ConnectX-3	xe-0/2/2 — Up 10G (VLAN 30)
QuarkyLab	onboard 1G	ge-0/0/24 — trunk (native 1 + VLAN 30)

2.4 OPNsense

OPNsense runs as VM 100 on pve2 (192.168.10.204). Version 25.1.12. Handles inter-VLAN routing, firewall, DHCP for all VLANs, and WAN. Serial console access: `qm terminal 100` from pve2.

3 Randy — SuperMicro CSE-219U

Randy — Commissioned 2026-06-22

[redacted]

3.1 Storage Layout

Device	Type	Purpose
/dev/sda (RAID-1)	185.8 GB SAS SSD x2	Proxmox OS boot mirror
18x Toshiba AL15SEB18EQ	1.636TB 10K SAS	datastore — 3x 6-wide RAIDZ2
4x Seagate ST2000NX0423	1.819TB SATA	datastore — 1x 4-wide RAIDZ2

3.1.1 RAID Controller — AVAGO 3108 MegaRAID

- Boot mirror: RAID-1 on slots 0:0 and 0:1 (Seagate ST200FM0053 SSDs)
- Data drives: JBOD mode enabled, all 22 data drives passed through to OS
- StorCLI installed at: `/opt/MegaRAID/storcli/storcli64`
- Symlink: `/usr/sbin/storcli64`

Listing 1: Useful StorCLI commands

```
# Show all drives and state
storcli64 /c0/eall/sall show

# Show controller info
```

```
storcli64 /c0 show

# Enable JBOD mode (if needed after reboot)
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod
```

3.1.2 ZFS Pool — datastore

Listing 2: ZFS pool layout

```
zpool status datastore
# pool: datastore
# state: ONLINE
#   raidz2-0: 6x Toshiba 1.636TB
#   raidz2-1: 6x Toshiba 1.636TB
#   raidz2-2: 6x Toshiba 1.636TB
#   raidz2-3: 4x Seagate ST2000NX0423 1.819TB
# 36.7TB raw / ~23TB usable (all drives in-pool; no spares)
```

3.2 Proxmox Backup Server

- PBS version: 4.2.2
- Web UI: `https://192.168.10.187:8007`
- Datastore: `datastore` at `/datastore`
- Fingerprint: stored in Vaultwarden (not published)
- Storage ID in cluster: `randy-pbs` (Shared: Yes, all nodes)

3.3 Networking

- **nic3** (Mellanox ConnectX-3): 10G UP → xe-0/2/0 on EX3400
- **Dual-homed (2026-07-02):** mgmt/corosync on VLAN 1 (192.168.10.187); NFS/PBS/egress on VLAN 30 (192.168.30.187)
- **nic0, nic1** (onboard Intel): unused
- vmbr0 bridges nic3 for management

3.3.1 Randy Commissioning Procedure (for rebuild reference)

Listing 3: Full Randy rebuild procedure

```
# 1. Boot IPMI BIOS -- hit Ctrl+H for MegaRAID WebBIOS
#    Create RAID-1 VD on two 185GB Seagate SSDs (slots 0:0, 0:1)

# 2. Install Proxmox VE 9.1 via USB
#    Target: /dev/sda (the RAID-1 VD, 185.78GiB, SMC3108)
#    Filesystem: ext4
#    Hostname: randy.netframe.local
#    IP: 192.168.10.187/24  GW: 192.168.10.1  DNS: 192.168.10.177
```

```
# 3. Post-install repo fix
echo "# disabled" > /etc/apt/sources.list.d/pve-enterprise.list
echo "# disabled" > /etc/apt/sources.list.d/ceph.list
echo "" > /etc/apt/sources.list.d/pve-enterprise.sources
echo "" > /etc/apt/sources.list.d/ceph.sources
echo "deb http://download.proxmox.com/debian/pve trixie pve-no-subscription" \
  > /etc/apt/sources.list.d/pve-no-subscription.list   # trixie (Debian 13/PVE 9); NOT bookworm
apt update && apt dist-upgrade -y

# 4. Join cluster
echo "192.168.10.204 pve2.netframe.local pve2" >> /etc/hosts
pvecm add 192.168.10.204 --use_ssh

# 5. Enable JBOD passthrough
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod

# 6. Create ZFS pool
zpool create -f -o ashift=12 datastore \
  raidz2 sdb sdd sde sdf sdh sdi \
  raidz2 sdj sdk sdl sdm sdn sdo \
  raidz2 sdp sdq sdr sds sdt sdu \
  raidz2 <4x Seagate ST2000NX0423>   # 4th vdev, added in the 2026-06-25 expansion

# 7. Install PBS
echo "deb http://download.proxmox.com/debian/pbs trixie pbs-no-subscription" \
  > /etc/apt/sources.list.d/pbs-no-subscription.list
apt update && apt install proxmox-backup-server -y

# 8. Create PBS datastore
proxmox-backup-manager datastore create datastore /datastore
```

4 QuarkyLab — Dell R730 #1

QuarkyLab

[redacted]

GPU stack (verified 2026-07-01): NVIDIA 550.163.01 on kernel 6.14.11-9-pve, pinned via `GRUB_DEFAULT` (6.17+ breaks the nvidia-550 DRM API). `nvidia-smi` reports 48 GB (driver-free Turing swap from the earlier RTX 6000).

Listing 4: Kernel pin for the NVIDIA 550 stack (QuarkyLab)

```
apt install pve-kernel-6.14.11-9-pve
```

```
# Pin as default in /etc/default/grub (GRUB_DEFAULT), then:
update-grub && reboot
# NVIDIA 550.163.01 driver already installed & verified -- do NOT
    upgrade the kernel
```

5 Jarvis — Dell R730 #2

Jarvis

[redacted]

Ollama serves Qwen2.5 72B Q4_K_M, GPU-backed across both cards (models on the `tank/models` ZFS dataset). FastAPI `llm_router.py` implements hybrid routing — local Ollama by default, escalating to the Claude API on an explicit `escalate` flag, a `model=claude-*` request, or local failure (Ollama exposes no logprobs, so routing is by flag/model/failure, not confidence).

6 Services

6.1 Running Services

Service	Host	VM/CT ID	IP	Notes
OPNsense	pve2	VM 100	192.168.10.1	Router/firewall
Pi-hole	pve1	LXC 103	192.168.10.177	DNS (Mac Mini, standalone)
Headscale	pve5	LXC 105	192.168.10.186	VPN, v0.29.1
Wazuh	QuarkyLab	VM 104	192.168.10.184	SIEM
step-ca	pve2	—	—	Internal CA
PBS	Randy	—	192.168.10.187:8007	Backup server

6.2 Headscale

Headscale v0.29.1 running as LXC 105 on pve5. Phase 1 (2026-06-22) migrated Ares (100.64.0.1), Randy (.2), pve5 (.3), pve4 (.4), pve3 (.5) and Jarvis (.6) to self-hosted. Phase 2 (QuarkyLab + the researcher's Mac) is pending.

6.3 Internal CA (step-ca)

step-ca 0.30.2 on pve2 issuing `*.netframe.local` TLS certificates.

7 Power Infrastructure

UPS	Feeds	Battery
Middle Atlantic UPS-OL2200R	R730s, Randy, DS4246	6x 12V 9Ah AGM in series (76.4V)
Tripp Lite SMART1500VA	EX3400, UniFi, small compute	Stock

PDU: APC AP7901 (ge-0/0/38 on EX3400).

8 Pending / To-Do

- ✓ VLAN activation — EX3400 trunk live, verified end-to-end (2026-06-25)
- ✓ QuarkyLab RTX 8000 + Jarvis 2x RTX 6000 installed & GPU-backed (2026-07)
- ✓ Backup schedules to randy-pbs (daily, 7d+4w retention)
- ✓ DS4246 — pool `bulk` built & online (2x 8-wide RAIDZ2, 41.3 TiB, 2026-07-08)
- ✓ Jellyfin on Randy (v10.11.11); RX 580 ROCm transcoding pending power cable
- ✓ UPS monitoring live (NUT → PeaNUT → Homepage/Grafana)
- ☐ Headscale Phase 2 — QuarkyLab + the researcher's Mac
- ☐ FreePBX VoIP (5x Cisco CP-8841 phones acquired)
- ☐ Cyberpunk monitoring dashboard — live API integration
- ☐ RKE2 / Kubernetes (Cilium, MetalLB, NVIDIA GPU Operator)

9 Quick Reference Commands

9.1 Cluster Health

```
# From any node
pvecm status
pvecm nodes

# PBS status
systemctl status proxmox-backup
proxmox-backup-manager datastore list

# ZFS pool health
zpool status datastore
zpool list
```

9.2 EX3400 Switch

```
ssh mason@192.168.10.50
# Always exit to > operational mode before show commands
# Paste one line at a time

show interfaces xe-0/2/0 brief    # Randy 10G
show interfaces xe-0/2/2 brief    # Jarvis 10G (ConnectX)
show ethernet-switching table      # MAC table
show lldp neighbors                # Connected devices
```

9.3 Randy Storage

```
ssh root@192.168.10.187

# Check all drives
storcli64 /c0/eall/sall show

# ZFS status
zpool status datastore
zfs list

# If drives go dark after reboot (JBOD mode reset)
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod
```